# Mathematical Notes: Rounded-Polygon Packing Model

**Corrected, tensorised, and annotated for the working physicist**

Derivations underlying `pyCudaPolygon`

Revised May 16, 2026

**Abstract**

This is a re-derivation of the rounded-polygon packing mathematics with two goals. First, every algebraic step is checked: six logical errors in the original notes are isolated and corrected, each flagged in a red box at the point of use. Second, the geometry is rewritten in index notation with the two-dimensional Levi-Civita symbol $\epsilon_{ij}$, which collapses the scattered cross-products, shoelace sums and rotation-by-90° operations into a handful of contractions. The physical content — intersection area as a soft energy, its analytic gradient, and constrained relaxation — is unchanged; only the bookkeeping is cleaner.

## Contents

## 1 Errata at a glance

The original notes are correct in their final results far more often than in their intermediate steps. The six items below are genuine logical errors (not typos); each is corrected in place later in the document, but they are collected here so a reader who only wants the audit can stop after this section.

1. **§3.2, tangent set-back.** The quadratic route is internally inconsistent. Imposing $\alpha = -\ell$ makes the $\hat{\boldsymbol{u}}$-component of $(\boldsymbol{z}-\boldsymbol{a}^-)$ vanish, so $|\boldsymbol{z}-\boldsymbol{a}^-|^2 = \beta^2 = \delta^2$ *immediately*. The notes instead write $|\boldsymbol{z}-\boldsymbol{a}^-|^2 = \alpha^2 + \beta^2 - 2\alpha\ell + \ell^2$ (wrong sign on the cross term) and conclude $\ell^2 + \beta^2 = \delta^2$, which does not follow. Carried through, their own algebra gives $\ell = \delta|c|/\sqrt{2(1+d)} = \delta\sin(\phi/2)$, contradicting the boxed answer $\delta\tan(\phi/2)$. The boxed answer is right; the derivation is not. See §4.1.

2. **§3.4, removed-corner area.** The first term must be $\frac{1}{2}\ell^2\sin\phi$ (the triangle $\boldsymbol{a}^-\boldsymbol{a}\,\boldsymbol{a}^+$ has included *interior* angle $\pi-\phi$), not $\frac{1}{2}\ell^2\tan(\phi/2)$. The supporting sentence also slips by a factor of two: $\sin\phi/\cos^2(\phi/2) = 2\tan(\phi/2)$, not $\tan(\phi/2)$. At $\phi = \pi/3$ the printed formula gives $0.0056\,\delta^2$ versus the true $0.0538\,\delta^2$. See §4.3.

3. **§4.4, arc integrand.** In the displayed line $\mathrm{d}x$ and $\mathrm{d}y$ are interchanged, which flips the sign of the $\delta^2\Delta\psi$ term and mispairs the Cartesian components. The boxed $\delta A_{\mathrm{arc}}$ is nevertheless correct because it is read off the geometric decomposition, not the botched integral. See §5.3.

4. **§8.2, velocity Verlet.** The position update double-counts the acceleration: $\boldsymbol{x}^n + \boldsymbol{v}^{n+1/2}\,\mathrm{d}t + \frac{1}{2}\boldsymbol{F}^n\,\mathrm{d}t^2 = \boldsymbol{x}^n + \boldsymbol{v}^n\,\mathrm{d}t + \boldsymbol{F}^n\,\mathrm{d}t^2$, not the stated $\boldsymbol{x}^n + \boldsymbol{v}^n\,\mathrm{d}t + \frac{1}{2}\boldsymbol{F}^n\,\mathrm{d}t^2$. The correct half-step form is simply $\boldsymbol{x}^{n+1} = \boldsymbol{x}^n + \boldsymbol{v}^{n+1/2}\,\mathrm{d}t$. See §9.

5. **§9.5, SLSQP range constraint.** "$\pi > \sum_k \phi_k > \pi$" is an empty set. For a closed simple polygon the exterior angles satisfy $\sum_k \phi_k = 2\pi$; the intended constraint is that equality (optionally a band around $2\pi$). See §10.

6. **§7.2, edge-gradient SVD.** With $G \in \mathbb{R}^{n\times 2n}$ one cannot have $U \in \mathbb{R}^{2n\times n}$ in $G = U\Sigma V^\top$; that is the SVD of $G^\top$. Stated as the SVD of $G^\top$, the projector identity $G^\top(GG^\top)^{-1}G = UU^\top$ is correct. See §8.

Two cosmetic points worth fixing while reading: the symbol "wrap" denotes the minimum-image map $x - \lfloor x + \frac{1}{2}\rfloor$ for *displacements* but the fold $x - \lfloor x \rfloor$ for *positions* (§8.2 step 2); these are different functions and should not share a name. And the estimate $\tan(\pi/N) \approx \pi/N$ used to bound the rounding radius (§10.3) is only accurate for large $N$ — harmless here since the exact inequality is checked, but it is not an identity.

## 2 Notation and tensor conventions

Latin indices $i, j, l, m \in \{1, 2\}$ label Cartesian components in the plane and the Einstein summation convention is used for them. Vertex labels are written as subscripts $a, b, \dots$ (so $x_a^i$ is component $i$ of vertex $a$); these are never summed implicitly. Because the metric is the identity, raising and lowering indices is trivial and used only for visual balance.

The single object that does all the geometric work is the two-dimensional *Levi-Civita symbol*

$$\epsilon_{ij} = -\epsilon_{ji}, \qquad \epsilon_{12} = +1, \qquad \epsilon_{11} = \epsilon_{22} = 0.$$

Its two contraction identities,

$$\epsilon_{ij}\epsilon_{lm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}, \qquad \epsilon_{ij}\epsilon_{lj} = \delta_{il}, \tag{1}$$

replace every trigonometric manipulation in the original notes. With it:

- **Scalar cross product:** $\boldsymbol{u} \times \boldsymbol{v} \equiv \epsilon_{ij} u^i v^j = u^1 v^2 - u^2 v^1$.
- **Dot product:** $\boldsymbol{u} \cdot \boldsymbol{v} = \delta_{ij} u^i v^j$.
- $+90°$ **(CCW) rotation:** $(\boldsymbol{u}^\perp)^i \equiv -\epsilon^{ij} u_j$, so $\boldsymbol{u}^\perp = (-u^2, u^1)$. This is the inward normal used below and matches the original notes' $\hat{\boldsymbol{n}} = (-\hat{u}_y, \hat{u}_x)$. Useful identities: $\boldsymbol{u}^\perp \cdot \boldsymbol{u} = 0$, $|\boldsymbol{u}^\perp| = |\boldsymbol{u}|$, $\boldsymbol{u} \times \boldsymbol{v} = \boldsymbol{u}^\perp \cdot \boldsymbol{v}$.

**Why bother.** Every "shoelace" sum, every signed area, and the entire 90°-rotation algebra in the original notes is one and the same operation: contraction with $\epsilon_{ij}$. Green's theorem, the corner geometry, the SHAKE Jacobian and the force projector all become contractions of $\delta_{ij}$ and $\epsilon_{ij}$, and the two identities (1) are the only trig you need.

Positions live in the periodic box $[0, 1)^2$. For *displacements* the minimum-image map is $\mathrm{mic}(x) = x - \lfloor x + \frac{1}{2} \rfloor \in [-\frac{1}{2}, \frac{1}{2})$, applied componentwise; for *positions* the fold is $\mathrm{fold}(x) = x - \lfloor x \rfloor \in [0, 1)$. (These are the two distinct maps the original notes both call "wrap".)

## 3 Polygon geometry

A backbone polygon $P$ has vertices $\boldsymbol{a}_0, \dots, \boldsymbol{a}_{N-1} \in [0, 1)^2$ traversed counterclockwise, with edge $E_a$ running from $\boldsymbol{a}_a$ to $\boldsymbol{a}_{a+1}$ (indices mod $N$). The minimum-imaged edge vectors are

$$u_a^i = \mathrm{mic}\big(a_a^i - a_{a-1}^i\big), \qquad v_a^i = \mathrm{mic}\big(a_{a+1}^i - a_a^i\big), \qquad \hat{\boldsymbol{u}}_a = \boldsymbol{u}_a/|\boldsymbol{u}_a|, \quad \hat{\boldsymbol{v}}_a = \boldsymbol{v}_a/|\boldsymbol{v}_a|.$$

### 3.1 Signed area

The shoelace formula is a single Levi-Civita contraction,

$$A_{\mathrm{bb}} = \tfrac{1}{2} \sum_{a=0}^{N-1} \epsilon_{ij}\, x_a^i\, x_{a+1}^j, \tag{2}$$

positive for CCW orientation, with all coordinates taken in one minimum-image frame anchored at $\boldsymbol{a}_0$. Differentiating (2) (each vertex appears in two terms) gives the area gradient that SHAKE needs,

$$g_a^i \;\equiv\; \frac{\partial A_{\mathrm{bb}}}{\partial x_a^i} = \tfrac{1}{2}\, \epsilon_{ij} \big(x_{a+1}^j - x_{a-1}^j\big), \tag{3}$$

which reproduces $\partial A/\partial x_a = \frac{1}{2}(y_{a+1} - y_{a-1})$ and $\partial A/\partial y_a = \frac{1}{2}(x_{a-1} - x_{a+1})$ on writing out the two components.

### 3.2 Turning angle

At vertex $a$ define the scalars

$$c_a = \epsilon_{ij}\hat{u}_a^i\hat{v}_a^j = \hat{\boldsymbol{u}}_a \times \hat{\boldsymbol{v}}_a, \qquad d_a = \delta_{ij}\hat{u}_a^i\hat{v}_a^j = \hat{\boldsymbol{u}}_a \cdot \hat{\boldsymbol{v}}_a,$$

so $c_a = \sin\phi_a$, $d_a = \cos\phi_a$ and the exterior (turning) angle is $\phi_a = \operatorname{atan2}(c_a, d_a)$. For a convex CCW polygon $\phi_a > 0$ and $\sum_a \phi_a = 2\pi$. The equilateral constraint enforced by SHAKE is $|\boldsymbol{v}_a| = L_a$ (equal edge lengths, not equal angles); the shape index is $\kappa = P/\sqrt{A}$ with $P = \sum_a L_a$ (circle $2\sqrt{\pi} \approx 3.545$, square 4).

## 4 The rounded corner

At each vertex the sharp corner is replaced by a circular arc of radius $\delta > 0$, tangent to both incident edges and curving *inward*. The boundary $\partial\tilde{P}$ alternates arcs $K_a$ and straight sections $E_a$. This is an inscribed rounding (it cuts the corner), not the outward Minkowski sum $P \oplus D(\delta)$.

### 4.1 Tangent set-back and arc centre

Drop the vertex index. Let $\hat{\boldsymbol{n}} = \hat{\boldsymbol{u}}^{\perp}$ be the inward unit normal to the incoming edge, so $\{\hat{\boldsymbol{u}}, \hat{\boldsymbol{n}}\}$ is an orthonormal frame. Resolve the outgoing direction in it:

$$\hat{\boldsymbol{v}} = d\,\hat{\boldsymbol{u}} + c\,\hat{\boldsymbol{n}}, \qquad d = \hat{\boldsymbol{u}} \cdot \hat{\boldsymbol{v}}, \quad c = \hat{\boldsymbol{u}} \times \hat{\boldsymbol{v}}, \quad c^2 + d^2 = 1. \tag{4}$$

Write the centre as $\boldsymbol{z} = \boldsymbol{a} + \alpha\,\hat{\boldsymbol{u}} + \beta\,\hat{\boldsymbol{n}}$. The two tangent points are $\boldsymbol{a}^- = \boldsymbol{a} - \ell\,\hat{\boldsymbol{u}}$ (incoming edge) and $\boldsymbol{a}^+ = \boldsymbol{a} + \ell\,\hat{\boldsymbol{v}}$ (outgoing edge).

**Tangency on the incoming edge.** The radius to $\boldsymbol{a}^-$ must be perpendicular to $\hat{\boldsymbol{u}}$, i.e. the $\hat{\boldsymbol{u}}$-component of $\boldsymbol{z} - \boldsymbol{a}^-$ must vanish:

$$\boldsymbol{z} - \boldsymbol{a}^- = (\alpha + \ell)\,\hat{\boldsymbol{u}} + \beta\,\hat{\boldsymbol{n}} \implies (\boldsymbol{z} - \boldsymbol{a}^-) \cdot \hat{\boldsymbol{u}} = \alpha + \ell = 0 \implies \boxed{\alpha = -\ell.}$$

**Tangency on the outgoing edge.** Using (4),

$$(\boldsymbol{z} - \boldsymbol{a}^+) \cdot \hat{\boldsymbol{v}} = \big(\alpha\,\hat{\boldsymbol{u}} + \beta\,\hat{\boldsymbol{n}} - \ell\,\hat{\boldsymbol{v}}\big) \cdot \hat{\boldsymbol{v}} = \alpha d + \beta c - \ell = 0.$$

With $\alpha = -\ell$ this gives

$$\beta = \frac{\ell(1+d)}{c}. \tag{5}$$

**Correction — §3.2: drop the spurious quadratic**

Because $\alpha = -\ell$ *kills* the $\hat{\boldsymbol{u}}$-component of $\boldsymbol{z} - \boldsymbol{a}^-$, the radius condition is just

$$|\boldsymbol{z} - \boldsymbol{a}^-|^2 = \left|\,\beta\,\hat{\boldsymbol{n}}\,\right|^2 = \beta^2 = \delta^2 \implies \beta = \delta \quad (\text{for } c > 0).$$

**No quadratic in $\ell$ arises.** The original notes write $|\boldsymbol{z} - \boldsymbol{a}^-|^2 = \alpha^2 + \beta^2 - 2\alpha\ell + \ell^2$ — the cross term has the wrong sign, since $\boldsymbol{z} - \boldsymbol{a}^- = (\alpha + \ell)\hat{\boldsymbol{u}} + \beta\hat{\boldsymbol{n}}$ gives $(\alpha + \ell)^2 + \beta^2$ — and then assert "$\Rightarrow \ell^2 + \beta^2 = \delta^2$," which does not follow from either sign. Propagating their expression yields $\ell = \delta|c|/\sqrt{2(1+d)} = \delta\sin(\phi/2)$, in direct contradiction with their own boxed $\delta\tan(\phi/2)$. The clean route below gives $\delta\tan(\phi/2)$ with no quadratic and no contradiction.

Setting $\beta = \delta$ in (5) and using $c = \sin\phi$, $d = \cos\phi$ with the half-angle identity $\sin\phi/(1+\cos\phi) = \tan(\phi/2)$:

$$\ell = \frac{\delta\, c}{1+d} = \delta\,\frac{\sin\phi}{1+\cos\phi} = \delta\tan\frac{\phi}{2}, \qquad \boldsymbol{z} = \boldsymbol{a} - \ell\,\hat{\boldsymbol{u}} + \delta\,\hat{\boldsymbol{n}}.$$

The centre can equivalently be written $\boldsymbol{z} = \boldsymbol{a} + \dfrac{\delta(\hat{\boldsymbol{v}} - \hat{\boldsymbol{u}})}{|c|}$: substituting $\hat{\boldsymbol{v}} - \hat{\boldsymbol{u}} = (d-1)\hat{\boldsymbol{u}} + c\,\hat{\boldsymbol{n}}$ and $\dfrac{d-1}{|c|} = -\dfrac{1-d}{|c|} = -\dfrac{c}{1+d} = -\tan(\phi/2)$ (using $1-d^2 = c^2$) returns $-\ell\,\hat{\boldsymbol{u}} + \delta\,\hat{\boldsymbol{n}}$ exactly. **Validity:** adjacent arcs do not collide iff $\ell_a + \ell_{a-1} < L_{a-1}$ for every edge; for equilateral edges a sufficient condition is $2\delta\tan(\phi_a/2) < L$ for all $a$, i.e. $\delta < L/\big(2\tan(\pi/N)\big)$ for a regular $N$-gon.

**Arc span.** The arc runs from $\boldsymbol{a}^-$ to $\boldsymbol{a}^+$ about $\boldsymbol{z}$. Its start angle is $\varphi_0 = \operatorname{atan2}\big(a_y^- - z_y,\ a_x^- - z_x\big)$ and it sweeps exactly the turning angle, $\Delta\phi = \phi$.

### 4.2 Straight section

The flat part of edge $a$ runs from $\boldsymbol{a}_a^+$ to $\boldsymbol{a}_{a+1}^-$, a segment of length $L_a - \ell_a - \ell_{a+1}$.

### 4.3 Area of the rounded polygon

The rounded area is $A_{\tilde{P}} = A_{\mathrm{bb}} - \sum_a \Delta A_a$, where $\Delta A_a$ is the area sliced off at vertex $a$ — the region bounded by the two tangent segments $\boldsymbol{a}^- \to \boldsymbol{a}$, $\boldsymbol{a} \to \boldsymbol{a}^+$ and the arc.

**Picture.** The removed piece is a triangle minus a circular cap. Triangle $\boldsymbol{a}^-\boldsymbol{a}\,\boldsymbol{a}^+$ has two sides of length $\ell$ meeting at the *interior* angle $\pi - \phi$, so its area is $\frac{1}{2}\ell^2\sin(\pi-\phi) = \frac{1}{2}\ell^2\sin\phi$. The arc bulges into that triangle; the slice that ends up *outside* the rounded body is the circular segment between chord $\boldsymbol{a}^-\boldsymbol{a}^+$ and the arc, of area $\frac{1}{2}\delta^2(\phi - \sin\phi)$.

**Correction — §3.4: $\sin\phi$, not $\tan(\phi/2)$, in the triangle term**

The triangle at the corner has included *interior* angle $\pi - \phi$, giving area $\frac{1}{2}\ell^2\sin\phi$. The original notes print $\frac{1}{2}\ell^2\tan(\phi/2)$ and "derive" it from $\frac{1}{2}\ell^2\sin\phi/\cos^2(\phi/2)$, but $\sin\phi/\cos^2(\phi/2) = 2\tan(\phi/2)$, so even their own expression equals $\ell^2\tan(\phi/2)$ (a stray factor of 2), and neither matches $\frac{1}{2}\ell^2\sin\phi$. Numerically at $\phi = \pi/3$ the printed formula yields $0.0056\,\delta^2$ while the correct value is $0.0538\,\delta^2$ — an order of magnitude, and even the leading small-$\phi$ behaviour has the wrong sign.

$$\Delta A_a = \underbrace{\tfrac{1}{2}\,\ell_a^2\sin\phi_a}_{\text{triangle } \boldsymbol{a}^-\boldsymbol{a}\,\boldsymbol{a}^+} - \underbrace{\tfrac{1}{2}\,\delta^2\big(\phi_a - \sin\phi_a\big)}_{\text{circular segment}} = \delta^2\left[\tan\frac{\phi_a}{2} - \frac{\phi_a}{2}\right].$$

The last equality (substitute $\ell = \delta\tan(\phi/2)$ and simplify with the half-angle identities) is the most useful closed form: it is manifestly $\geq 0$, vanishes as $\phi^3/12$ for small turns, and depends only on $\delta$ and $\phi$. Equivalently $\Delta A_a = \ell_a\delta - \frac{1}{2}\delta^2\phi_a$ via the kite-minus-sector decomposition — a handy independent check ($\ell\delta = \delta^2\tan(\phi/2)$).

## 5 Intersection area as the energy

For two rounded polygons $\tilde{A}, \tilde{B}$ the pair energy is the overlap area $E_{AB} = \mathrm{Area}(\tilde{A} \cap \tilde{B})$ and the total energy is $E = \sum_{A<B} E_{AB}$.

### 5.1 Green's theorem, tensorially

Green's theorem in shoelace form is, for a region $\Omega$ with CCW boundary,

$$\mathrm{Area}(\Omega) = \tfrac{1}{2} \oint_{\partial\Omega} \epsilon_{ij}\, x^i \, \mathrm{d}x^j. \tag{6}$$

The key trick (used by the kernel to avoid clipping) is locality: applying (6) to $\tilde{A} \cap \tilde{B}$, only the parts of $\partial\tilde{A}$ that lie *inside* $\tilde{B}$ contribute. So one walks each feature of $\partial\tilde{A}$ (arc $K_i$ or edge $E_i$), finds the sub-intervals inside $\tilde{B}$, and adds their signed area. The mirror contributions from $\partial\tilde{B}$ inside $\tilde{A}$ are picked up when the roles swap; `atomicAdd` sums both halves.

### 5.2 Straight-edge contribution

For an inside sub-interval with endpoints $\boldsymbol{X}_0, \boldsymbol{X}_1$ on a straight feature, parametrise $\boldsymbol{X}(s) = (1 - s)\boldsymbol{X}_0 + s\boldsymbol{X}_1$. Then with (1),

$$\tfrac{1}{2} \int_0^1 \epsilon_{ij} X^i \, \mathrm{d}X^j = \tfrac{1}{2}\, \epsilon_{ij} \left[ \int_0^1 \big((1-s)X_0 + sX_1\big)^i \big(X_1 - X_0\big)^j \, \mathrm{d}s \right] = \tfrac{1}{2}\, \epsilon_{ij} X_0^i X_1^j,$$

the $X_0X_0$ and $X_1X_1$ pieces vanishing by antisymmetry of $\epsilon_{ij}$.

$$\delta A_{\mathrm{edge}} = \tfrac{1}{2}\, \epsilon_{ij}\, X_0^i X_1^j = \tfrac{1}{2}\big(X_{0x}X_{1y} - X_{0y}X_{1x}\big),$$

matching the kernel line `localEnergy += 0.5*(X0x*X1y - X0y*X1x)`.

### 5.3 Arc contribution

An arc of radius $\delta$ about $\boldsymbol{z}$ is $\boldsymbol{X}(\varphi) = \boldsymbol{z} + \delta(\cos\varphi, \sin\varphi)$. The differentials are

$$\mathrm{d}x = -\delta \sin\varphi \, \mathrm{d}\varphi, \qquad \mathrm{d}y = +\delta \cos\varphi \, \mathrm{d}\varphi,$$

so the Green integrand is

$$\epsilon_{ij} X^i \, \mathrm{d}X^j = x \, \mathrm{d}y - y \, \mathrm{d}x = \delta\big(z_x \cos\varphi + z_y \sin\varphi\big) \, \mathrm{d}\varphi + \delta^2 \, \mathrm{d}\varphi.$$

**Correction — §4.4: $\mathrm{d}x$ and $\mathrm{d}y$ were swapped**

The original notes display $x \, \mathrm{d}y - y \, \mathrm{d}x = (z_x + \delta\cos\varphi)(-\delta \sin\varphi \, \mathrm{d}\varphi) - (z_y + \delta \sin\varphi)(\delta\cos\varphi \, \mathrm{d}\varphi)$, i.e. they used $\mathrm{d}y = -\delta\sin\varphi \, \mathrm{d}\varphi$ and $\mathrm{d}x = +\delta\cos\varphi \, \mathrm{d}\varphi$ — the two are interchanged. This flips the sign of the $\delta^2\Delta\psi$ term and pairs $z_x$ with $X_{1x}$ instead of $X_{1y}$. The boxed final $\delta A_{\mathrm{arc}}$ survives only because it is read off the geometric decomposition, not this integral.

Integrating from $\varphi_0$ to $\varphi_1$ and using $\delta\cos\varphi_{0/1} = X_{0/1,x} - z_x$, $\delta\sin\varphi_{0/1} = X_{0/1,y} - z_y$,

$$\int_{\varphi_0}^{\varphi_1} \big(x \, \mathrm{d}y - y \, \mathrm{d}x\big) = z_x\big(X_{1y} - X_{0y}\big) - z_y\big(X_{1x} - X_{0x}\big) + \delta^2\Delta\psi, \qquad \Delta\psi = \varphi_1 - \varphi_0.$$

Halving and regrouping into "triangle from the origin + circular segment" gives the result the kernel actually uses:

$$\delta A_{\mathrm{arc}} = \tfrac{1}{2}\,\epsilon_{ij} X_0^i X_1^j + \tfrac{1}{2}\,\delta^2\big(\Delta\psi - \sin\Delta\psi\big) = \tfrac{1}{2}\big(X_{0x}X_{1y} - X_{0y}X_{1x}\big) + \tfrac{1}{2}\delta^2\big(\Delta\psi - \sin\Delta\psi\big).$$

This matches `localEnergy += 0.5*(X0x*X1y - X0y*X1x) + 0.5*delta*delta*(dpsi - sin(dpsi))`. A quick sanity check: a quarter circle of unit radius centred at the origin has $\boldsymbol{X}_0 = (1,0)$, $\boldsymbol{X}_1 = (0,1)$, $\Delta\psi = \pi/2$, giving $\frac{1}{2}(1) + \frac{1}{2}(\pi/2 - 1) = \pi/4$, the area of the quarter disk. ✓

### 5.4 Point-in-smoothed-polygon and feature intersections

Inside/outside is decided by a horizontal ray cast (`pipSmoothed`): count crossings of $\partial\tilde{B}$; odd means inside. Straight features use the standard ray–segment test; for an arc the ray $y = p_y$ meets the circle at $x = z_x \pm \sqrt{\delta^2 - (p_y - z_y)^2}$, and each root is kept only if it lies on the arc's angular range and has $x > p_x$. Feature intersections reduce to one quadratic or one linear solve each:

- *Arc–edge:* with $\boldsymbol{r}(t) = \boldsymbol{p} + t\boldsymbol{e}$, $\boldsymbol{f} = \boldsymbol{p} - \boldsymbol{z}$, the circle equation gives $|\boldsymbol{e}|^2 t^2 + 2(\boldsymbol{f}\cdot\boldsymbol{e})t + |\boldsymbol{f}|^2 - \delta^2 = 0$.
- *Arc–arc:* along the line of centres, offset $a_2 = (d^2 + \delta_A^2 - \delta_B^2)/(2d)$ and $h = \sqrt{\max(0, \delta_A^2 - a_2^2)}$, $\boldsymbol{X} = \boldsymbol{z}_A + a_2\hat{\boldsymbol{s}} \pm h\,\hat{\boldsymbol{s}}^{\perp}$.
- *Edge–edge:* $t = \dfrac{(\boldsymbol{r} - \boldsymbol{p})\times\boldsymbol{e}}{\boldsymbol{d}\times\boldsymbol{e}}$, $s = \dfrac{(\boldsymbol{r} - \boldsymbol{p})\times\boldsymbol{d}}{\boldsymbol{d}\times\boldsymbol{e}}$, with the cross product read as $\epsilon_{ij}$.

All three are algebraically sound in the original notes.

## 6 Force as the gradient of intersection area

The force on vertex $a$ is $F_a^i = -\partial E/\partial x_a^i$. Because both the integrand *and* the integration limits depend on the vertices, the derivative needs the Leibniz rule.

### 6.1 Leibniz rule for a signed-area contribution

For a feature $\gamma(t;\theta)$ with the contribution $I = \frac{1}{2}\int_{t_0}^{t_1} \epsilon_{ij}\gamma^i\dot{\gamma}^j\,\mathrm{d}t$,

$$\frac{\partial I}{\partial\theta} = \underbrace{\tfrac{1}{2}\int_{t_0}^{t_1} \frac{\partial}{\partial\theta}\big(\epsilon_{ij}\gamma^i\dot{\gamma}^j\big)\,\mathrm{d}t}_{\text{direct term}} \;+\; \underbrace{\Big[\tfrac{1}{2}\,\epsilon_{ij}\gamma^i\dot{\gamma}^j\Big]_{t_1}\frac{\partial t_1}{\partial\theta} - \Big[\tfrac{1}{2}\,\epsilon_{ij}\gamma^i\dot{\gamma}^j\Big]_{t_0}\frac{\partial t_0}{\partial\theta}}_{\text{boundary (limit) term}}. \tag{7}$$

The direct term is how the curve itself moves with the vertices; the boundary term is how the in/out crossing points slide.

### 6.2 Straight edge

For $E_i$ from $\boldsymbol{P} = \boldsymbol{a}_i^+$ to $\boldsymbol{Q} = \boldsymbol{a}_{i+1}^-$ with $I = \frac{1}{2}\epsilon_{ij}X_0^iX_1^j$ and $\boldsymbol{X}_{0,1} = \gamma(t_{0,1})$, $\gamma(t) = (1-t)\boldsymbol{P} + t\boldsymbol{Q}$, the direct term is linear in the barycentric weights:

$$\frac{\partial I}{\partial\boldsymbol{P}}\Big|_{\mathrm{dir}} = (1-t_0)\,\boldsymbol{f}_0 + (1-t_1)\,\boldsymbol{f}_1, \qquad \frac{\partial I}{\partial\boldsymbol{Q}}\Big|_{\mathrm{dir}} = t_0\,\boldsymbol{f}_0 + t_1\,\boldsymbol{f}_1,$$

with $f_0^i = \frac{1}{2}\epsilon_{ij}X_1^j$ and $f_1^i = -\frac{1}{2}\epsilon_{ij}X_0^j$ (the "$X^\perp$" vectors). For an edge–edge crossing at $t_0 = (\boldsymbol{f}\times\boldsymbol{e}_B)/(\boldsymbol{e}_A\times\boldsymbol{e}_B)$ the limit term uses

$$\frac{\partial t_0}{\partial \boldsymbol{P}} = \frac{-\boldsymbol{e}_B^\perp + t_0\,\boldsymbol{e}_A^\perp}{\boldsymbol{e}_A\times\boldsymbol{e}_B}, \qquad \frac{\partial I}{\partial t_0} = \tfrac{1}{2}\big(\boldsymbol{e}_A\times\boldsymbol{X}_1\big),$$

giving the "$\beta$-correction" contribution $\frac{1}{2}(\boldsymbol{e}_A\times\boldsymbol{X}_1)\,\partial t_0/\partial\boldsymbol{P}$ (and similarly for $t_1$). Here $\boldsymbol{e}_A = \boldsymbol{Q}-\boldsymbol{P}$ and $\boldsymbol{v}^\perp = \epsilon^{\cdot}{}_{j}v^j$ is the $90°$ rotation.

### 6.3 Arc segment and the arc-geometry chain rule

For $K_i$ the contribution is $I = \frac{1}{2}\epsilon_{ij}X_0^iX_1^j + \frac{1}{2}\delta^2(\Delta\psi - \sin\Delta\psi)$ with $\Delta\psi = (\tau_1-\tau_0)\Delta\phi_i$. The arc moves because $\boldsymbol{z}_i$, $\varphi_{0,i}$ and $\Delta\phi_i$ each depend on the three backbone vertices $\boldsymbol{a}_{i-1},\boldsymbol{a}_i,\boldsymbol{a}_{i+1}$. The routine `arcGeomGrad` takes the sensitivities $\partial E/\partial\boldsymbol{a}_i^-$, $\partial E/\partial\boldsymbol{a}_i^+$, $\partial E/\partial\Delta\phi_i$ and propagates them by the chain rule

$$\frac{\partial E}{\partial\boldsymbol{a}_{i-1}} = \frac{\partial E}{\partial\boldsymbol{a}_i^-}\frac{\partial\boldsymbol{a}_i^-}{\partial\boldsymbol{a}_{i-1}} + \frac{\partial E}{\partial\Delta\phi_i}\frac{\partial\Delta\phi_i}{\partial\boldsymbol{a}_{i-1}},$$

and analogously for $\boldsymbol{a}_i$ (which receives both tangent-point terms) and $\boldsymbol{a}_{i+1}$. The needed partials are clean in index form. With $\ell = \delta c/(1+d)$, $c = \hat{\boldsymbol{u}}\times\hat{\boldsymbol{v}}$, $d = \hat{\boldsymbol{u}}\cdot\hat{\boldsymbol{v}}$,

$$\frac{\partial\ell}{\partial\hat{u}^i} = \frac{1}{1+d}\Big(\operatorname{sgn}(c)\,\delta\,(\hat{\boldsymbol{v}}^\perp)_i - \ell\,\hat{v}_i\Big), \qquad \frac{\partial\,\Delta\phi}{\partial\hat{u}^i} = \epsilon_{ij}\hat{v}^j\,d - \hat{v}_i\,c \quad (\text{since } c^2{+}d^2{=}1),$$

and the normalisation derivative is the **projection tensor**

$$\frac{\partial\hat{u}^j}{\partial x_i^l} = \pm\frac{P^{jl}}{|\boldsymbol{u}|}, \qquad \boxed{P^{jl} = \delta^{jl} - \hat{u}^j\hat{u}^l} \quad (+\text{ for }\boldsymbol{a}_i,\ -\text{ for }\boldsymbol{a}_{i-1}), \tag{8}$$

the operator that strips the component along $\hat{\boldsymbol{u}}$ (consistent with $|\hat{\boldsymbol{u}}| = 1$). Assembling, the four vertex forces are

$$\boldsymbol{f}_{i-1} = -\frac{1}{|\boldsymbol{u}|}\,P\cdot\frac{\partial E}{\partial\hat{\boldsymbol{u}}}, \quad \boldsymbol{f}_{i+1} = \frac{1}{|\boldsymbol{v}|}\,P\cdot\frac{\partial E}{\partial\hat{\boldsymbol{v}}}, \quad \boldsymbol{f}_i = \boldsymbol{f}^{a-} + \boldsymbol{f}^{a+} + \frac{1}{|\boldsymbol{u}|}P\cdot\frac{\partial E}{\partial\hat{\boldsymbol{u}}} - \frac{1}{|\boldsymbol{v}|}P\cdot\frac{\partial E}{\partial\hat{\boldsymbol{v}}},$$

where $P^{jl} = \delta^{jl} - \hat{u}^j\hat{u}^l$ (resp. with $\hat{\boldsymbol{v}}$) is exactly (8). These partials are correct in the original notes; only the upstream $\ell$-derivation (§4.1) needed repair, and the correct $\ell = \delta\tan(\phi/2)$ is the form already used here.

## 7 SHAKE constraint projection

### 7.1 Constraints and Newton step

A polygon with $n$ vertices carries $n+1$ holonomic constraints,

$$C_k(\boldsymbol{x}) = |\boldsymbol{x}_{k+1}-\boldsymbol{x}_k| - L_k = 0 \ \ (k=0,\dots,n-1), \qquad C_n(\boldsymbol{x}) = A(\boldsymbol{x}) - A_0 = 0.$$

With Jacobian $J\in\mathbb{R}^{(n+1)\times 2n}$, $J_k = \nabla C_k$, one Newton step linearises $C(\boldsymbol{x}+\Delta\boldsymbol{x}) \approx C + J\Delta\boldsymbol{x} = 0$ with $\Delta\boldsymbol{x} = -J^\top\boldsymbol{\lambda}$, giving

$$\big(JJ^\top\big)\boldsymbol{\lambda} = C(\boldsymbol{x}), \qquad \Delta\boldsymbol{x} = -J^\top\boldsymbol{\lambda}.$$

### 7.2 Jacobian and Gram matrix

With unit edge $\hat{e}_k = (\boldsymbol{x}_{k+1} - \boldsymbol{x}_k)/|\boldsymbol{x}_{k+1} - \boldsymbol{x}_k|$,

$$\nabla_{\boldsymbol{x}_k} C_k = -\hat{e}_k, \qquad \nabla_{\boldsymbol{x}_{k+1}} C_k = +\hat{e}_k, \qquad \nabla_{\boldsymbol{x}_k} C_n = g_k \text{ from (3).}$$

The Gram matrix $M = JJ^\top$, $M_{kl} = \nabla C_k \cdot \nabla C_l$, is sparse:

$$M_{kk} = 2, \quad M_{k,k\pm1} = -\hat{e}_k \cdot \hat{e}_{k\pm1}, \quad M_{kj} = 0 \ (|k-j| > 1 \bmod n),$$
$$M_{n,k} = M_{k,n} = \hat{e}_k \cdot \big(g_{k+1} - g_k\big), \qquad M_{n,n} = \sum_k |g_k|^2.$$

(The edge–edge diagonal is 2 because $\nabla C_k$ carries $-\hat{e}_k$ and $+\hat{e}_k$; adjacent edges share exactly one vertex.) The $(n{+}1) \times (n{+}1)$ system $M\boldsymbol{\lambda} = C$ is solved by unpivoted LU (well conditioned for equilateral polygons), and the per-vertex update is

$$\Delta \boldsymbol{x}_k = \lambda_k \, \hat{e}_k - \lambda_{k-1} \, \hat{e}_{k-1} - \lambda_n \, g_k.$$

Convergence is $\max_k |C_k| < \text{tol} = 10^{-15}$; on failure the step count is decadally increased, and a final failure rejects the FIRE step. All of §6 is algebraically correct.

## 8 Force projection

After computing the overlap force $\boldsymbol{F}$ we remove its component in the constraint subspace so FIRE moves along the constraint manifold. The orthogonal projector is

$$\boldsymbol{F}_\perp = \boldsymbol{F} - J^\top \big(JJ^\top\big)^{-1} J \, \boldsymbol{F}.$$

Build the edge-gradient matrix $G$ whose row $k$ is $\nabla C_k$ for the $k$-th edge constraint, so $G \in \mathbb{R}^{n \times 2n}$ with nonzeros $G_{k,2k} = -\hat{e}_{k,x}$, $G_{k,2k+1} = -\hat{e}_{k,y}$, $G_{k,2(k+1)} = +\hat{e}_{k,x}$, $G_{k,2(k+1)+1} = +\hat{e}_{k,y}$.

**Correction — §7.2: it is the SVD of $G^\top$**

The constraint subspace lives in $\mathbb{R}^{2n}$; an orthonormal basis for it is the set of left singular vectors of $G^\top \in \mathbb{R}^{2n \times n}$, written $G^\top = U\Sigma V^\top$ with $U \in \mathbb{R}^{2n\times n}$, $V \in \mathbb{R}^{n \times n}$. (The original notes attach $U \in \mathbb{R}^{2n \times n}$ to $G = U\Sigma V^\top$ with $G \in \mathbb{R}^{n \times 2n}$, which is dimensionally impossible — that $U$ belongs to $G^\top$.) The columns $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n$ then span the row space of $G$.

For full row-rank $G$ the projector identity $G^\top(GG^\top)^{-1}G = UU^\top$ holds, so the edge part of the projection is

$$\boldsymbol{F}_{\perp,\text{edge}} = \boldsymbol{F} - UU^\top \boldsymbol{F} = \boldsymbol{F} - \sum_{j=1}^{n} (\boldsymbol{u}_j \cdot \boldsymbol{F}) \, \boldsymbol{u}_j.$$

The area gradient $\boldsymbol{g}_A = \nabla A$ is not generally in $\operatorname{span}\{\boldsymbol{u}_j\}$, so Gram–Schmidt it out and normalise,

$$\tilde{\boldsymbol{g}}_A = \boldsymbol{g}_A - \sum_{j=1}^{n} (\boldsymbol{u}_j \cdot \boldsymbol{g}_A) \, \boldsymbol{u}_j, \qquad \hat{\boldsymbol{q}}_A = \tilde{\boldsymbol{g}}_A / |\tilde{\boldsymbol{g}}_A|,$$

and finally project out both blocks,

$$\boxed{\boldsymbol{F}_\perp = \boldsymbol{F} - \sum_{j=1}^{n} (\boldsymbol{u}_j \cdot \boldsymbol{F}) \, \boldsymbol{u}_j - (\hat{\boldsymbol{q}}_A \cdot \boldsymbol{F}) \, \hat{\boldsymbol{q}}_A}$$

which has zero component along any constraint gradient.

# 9 FIRE minimisation

FIRE (Bitzek *et al.*, 2006) is a velocity-Verlet integrator periodically steered so the velocity points downhill.

## 9.1 Velocity Verlet (corrected)

**Correction — §8.2: the position update double-counted the acceleration**

The original notes write $\boldsymbol{x}^{n+1} = \boldsymbol{x}^n + \boldsymbol{v}^{n+1/2}\,\mathrm{d}t + \frac{1}{2}\boldsymbol{F}^n\,\mathrm{d}t^2$ and equate it to $\boldsymbol{x}^n + \boldsymbol{v}^n\,\mathrm{d}t + \frac{1}{2}\boldsymbol{F}^n\,\mathrm{d}t^2$. But substituting $\boldsymbol{v}^{n+1/2} = \boldsymbol{v}^n + \frac{1}{2}\boldsymbol{F}^n\,\mathrm{d}t$ gives $\boldsymbol{x}^n + \boldsymbol{v}^n\,\mathrm{d}t + \boldsymbol{F}^n\,\mathrm{d}t^2$ — the acceleration appears *twice*. The correct half-kick form carries no extra $\frac{1}{2}\boldsymbol{F}\,\mathrm{d}t^2$:

$$\boldsymbol{v}^{n+1/2} = \boldsymbol{v}^n + \tfrac{1}{2}\boldsymbol{F}^n\,\mathrm{d}t, \qquad \boxed{\boldsymbol{x}^{n+1} = \boldsymbol{x}^n + \boldsymbol{v}^{n+1/2}\,\mathrm{d}t} \quad \big(= \boldsymbol{x}^n + \boldsymbol{v}^n\,\mathrm{d}t + \tfrac{1}{2}\boldsymbol{F}^n\,\mathrm{d}t^2\big).$$

The remaining steps are correct: fold $\boldsymbol{x}^{n+1}$ into $[0,1)^2$; SHAKE-project onto the manifold; rederive the velocity from the realised displacement, $\boldsymbol{v}^{n+1/2}_{\text{rederived}} = \mathrm{mic}\big(\boldsymbol{x}^{n+1}_{\text{SHAKE}} - \boldsymbol{x}^n\big)/\,\mathrm{d}t$; recompute and project $\boldsymbol{F}^{n+1}$; then the second half-kick $\boldsymbol{v}^{n+1} = \boldsymbol{v}^{n+1/2}_{\text{rederived}} + \frac{1}{2}\boldsymbol{F}^{n+1}\,\mathrm{d}t$.

## 9.2 Steering

With power $\Pi = \boldsymbol{F}^{n+1}\cdot\boldsymbol{v}^{n+1}$: if $\Pi > 0$ bend the velocity toward the force, $\boldsymbol{v} \leftarrow (1-\alpha)\boldsymbol{v} + \alpha\,|\boldsymbol{v}|\,\hat{\boldsymbol{F}}$, and after $n_{\min}$ consecutive uphill-free steps grow the step, $\mathrm{d}t \leftarrow \min(\mathrm{d}t\,f_{\text{inc}},\,\mathrm{d}t_{\max})$, $\alpha \leftarrow \alpha f_\alpha$. If $\Pi \le 0$ zero the velocity, shrink $\mathrm{d}t \leftarrow \mathrm{d}t\,f_{\text{dec}}$, reset $\alpha \leftarrow \alpha_0$. The defaults ($\mathrm{d}t_0 = 10^{-3}$, $\mathrm{d}t_{\max} = 5\times10^{-2}$, $\alpha_0 = 0.1$, $f_\alpha = 0.99$, $f_{\text{inc}} = 1.1$, $f_{\text{dec}} = 0.5$, $n_{\min} = 5$) and the dense-packing bound $\mathrm{d}t_{\max} \lesssim \sqrt{\varepsilon L_{\min}/F_{\max}}$ (from $F_{\max}\,\mathrm{d}t^2_{\max} \lesssim \varepsilon L_{\min}$) are consistent.

# 10 Shape generation via SLSQP

Generate a random equilateral $N$-gon with shape index $\kappa$ and area $A_{\text{tgt}}$. Parametrise by the $N-1$ free turning angles $\boldsymbol{\phi} = (\phi_0,\dots,\phi_{N-2})$, build $\theta_k = \sum_{j<k}\phi_j$, $\hat{e}_k = (\cos\theta_k, \sin\theta_k)$, $\boldsymbol{v}_0 = 0$, $\boldsymbol{v}_{k+1} = \boldsymbol{v}_k + \hat{e}_k$ (unit edges, $L_k = 1$). Closure is $\sum_k \hat{e}_k = 0$ (two equality constraints). With unit edges $P = N$, so $\kappa = N/\sqrt{A}$ and $\kappa = \kappa_0$ becomes the single equality $A(\boldsymbol{\phi}) = (N/\kappa_0)^2 =: A_{\text{unit}}$.

**Correction — §9.5: "$\pi > \sum\phi_k > \pi$" is empty**

For a closed simple polygon the exterior angles sum to $2\pi$, so the intended turning-budget constraint is the equality $\sum_k \phi_k = 2\pi$ (or a small band around $2\pi$ if a soft version is wanted) — never $\pi > \sum_k \phi_k > \pi$, which admits no $\boldsymbol{\phi}$ at all.

$$\begin{aligned} \min_{\boldsymbol{\phi}} \quad & \tfrac{1}{2}|\boldsymbol{\phi} - \boldsymbol{\phi}_0|^2 \\ \text{s.t.} \quad & \sum_k \cos\theta_k = 0, \quad \sum_k \sin\theta_k = 0, \quad A(\boldsymbol{\phi}) = A_{\text{unit}}, \\ & \sum_k \phi_k = 2\pi, \qquad \phi_k \in (10^{-4},\, \pi - 10^{-4}). \end{aligned}$$

$\boldsymbol{\phi}_0$ is drawn from a Dirichlet distribution scaled to $[0, 2\pi]$ (random convex-ish reference). On success rescale to the target area, $\boldsymbol{v}_k \leftarrow \sqrt{A_{\mathrm{tgt}}/A_{\mathrm{unit}}}\,(\boldsymbol{v}_k - \bar{\boldsymbol{v}})$, and apply a random rotation to remove orientation bias.

# 11 Bidisperse packing, PBC, and computational flow

**Target areas.** With $N_L = N_S = N_{\mathrm{poly}}/2$ and equal shape index, $P_L = \mathrm{ratio} \cdot P_S \Rightarrow A_L = \mathrm{ratio}^2 A_S$. Substituting into the packing fraction $\phi = N_L A_L + N_S A_S = A_S(N_L\,\mathrm{ratio}^2 + N_S)$ gives

$$A_S = \frac{\phi}{N_L\,\mathrm{ratio}^2 + N_S}, \qquad A_L = \mathrm{ratio}^2 A_S.$$

**Target edge length and rounding.** From $P = NL = \kappa\sqrt{A}$, $L = \kappa\sqrt{A}/N$, and $\delta = 0.15\,L_S = 0.15\,\kappa\sqrt{A_S}/N$. The validity bound $\delta < L/\big(2\tan(\pi/N)\big)$ is checked exactly; the companion estimate $\tan(\pi/N) \approx \pi/N$ is a large-$N$ approximation, not an identity, and is comfortably satisfied here. For the two-polygon test $A_L = 0.8/64 = 0.012500$ and $A_S = A_L/\mathrm{ratio}^2$; with $\mathrm{ratio}^2 = 1.96$, $A_S \approx 0.006378$. ✓

**Periodic boundary conditions.** All positions lie in $[0, 1)^2$. Displacements use $\mathrm{mic}(x) = x - \lfloor x + \frac{1}{2} \rfloor$ componentwise. For the $A$–$B$ interaction, $B$ is placed in one consistent image, $\boldsymbol{b}_0^{\mathrm{MIC}} = \boldsymbol{a}_i + \mathrm{mic}(\boldsymbol{b}_0 - \boldsymbol{a}_i)$, $\boldsymbol{b}_j^{\mathrm{MIC}} = \boldsymbol{b}_0^{\mathrm{MIC}} + (\boldsymbol{b}_j - \boldsymbol{b}_0)$, so no edge straddles the wrap seam.

**Computational flow.** (1) SLSQP-generate $N_{\mathrm{poly}}$ equilateral polygons; place on a jittered grid; build neighbour cells. (2) For each vertex of each polygon, in parallel: compute arc geometry, gather neighbours in the 3×3 cell block, clip each arc/edge against every neighbouring $\tilde{B}$, integrate the signed area of inside sub-intervals, and accumulate forces on $\{i{-}1, i, i{+}1, i{+}2\}$ via the `arcGeomGrad` chain rule (and onto $B$'s vertices). (3) Project the force (SVD of $G^\top$ + Gram–Schmidt of $\boldsymbol{g}_A$). (4) FIRE step: half-kick → corrected position update → SHAKE → velocity rederivation → force → second half-kick → steering. (5) Iterate until $\|\boldsymbol{F}_\perp\|_\infty < F_{\mathrm{tol}}$.

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**Summary of corrections.** §3.2 (spurious quadratic; clean route gives $\ell = \delta\tan(\phi/2)$), §3.4 ($\Delta A = \frac{1}{2}\ell^2 \sin\phi - \frac{1}{2}\delta^2(\phi - \sin\phi)$), §4.4 ( $\mathrm{d}x \leftrightarrow \mathrm{d}y$ swap), §8.2 (double-counted acceleration), §9.5 (empty range constraint), §7.2 (SVD belongs to $G^\top$). Every final boxed formula in the original is correct *except* the removed-corner area in §3.4; the others had sound results reached by unsound steps.